HDOC Day-44

Write a menu-driven C program to perform the following operations using separate functions:

- Reverse each word in a sentence without changing the word order. Input: What is special?, Output: tahw si ?laiceps
- Print the initials of a name. E.g. Input: Alexander Grahambell, Output: A.G.

Algorithm:

Operation 1: Reverse Each Word Without Changing Word Order

Example: What is special? → tahW si ?laiceps

1. Start.
2. Input a sentence from the user.
3. Display a menu with the options:
    - Reverse each word.
    - Print initials of a name.
    - Exit.
4. If the user selects option 1:
    - Read the sentence character by character.
    - Identify each word using spaces as separators.
    - Reverse the characters of each word.
    - Keep the original order of words unchanged.
    - Display the modified sentence.
5. If the user selects option 2, execute the initials function.
6. If the user selects option 3, terminate the program.
7. Stop.

Operation 2: Print Initials of a Name

Example: Alexander Graham Bell → A.G.B.

1. Start.
2. Input the full name.

3. Read the first character of the name and print it in uppercase.
4. Whenever a space is encountered, check whether the next word exists.
5. If another word exists, print its first character followed by a dot.
6. Display the initials.
7. Stop.

2. C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX 200

// Function to reverse each word
void reverseWords(char str[])
{
   int i, start = 0, end, len;
   char temp;

   len = strlen(str);

   for (i = 0; i <= len; i++)
   {
      if (str[i] == ' ' || str[i] == '\0')
      {
         end = i - 1;

         while (start < end)
         {
```

```
            temp = str[start];
            str[start] = str[end];
            str[end] = temp;

            start++;
            end--;
        }

        start = i + 1;
    }
  }

  printf("Result: %s\n", str);
}

// Function to print initials
void printInitials(char name[])
{
  int i;

  printf("Initials: ");

  for (i = 0; name[i] != '\0'; i++)
  {
    if (i == 0 && name[i] != ' ')
      printf("%c.", toupper(name[i]));

    else if (name[i] == ' ' &&
```

```
            name[i + 1] != '\0' &&
            name[i + 1] != ' ')
        printf("%c.", toupper(name[i + 1]));
    }

    printf("\n");
}

int main()
{
    int choice;
    char str[MAX];

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Reverse each word\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        getchar();  // Remove newline left by scanf

        switch (choice)
        {
            case 1:
                printf("Enter a sentence: ");
```

```
            fgets(str, MAX, stdin);
            str[strcspn(str, "\n")] = '\0';

            reverseWords(str);
            break;

        case 2:
            printf("Enter full name: ");
            fgets(str, MAX, stdin);
            str[strcspn(str, "\n")] = '\0';

            printInitials(str);
            break;

        case 3:
            printf("Exiting program.\n");
            break;

        default:
            printf("Invalid choice!\n");
    }

  } while (choice != 3);

  return 0;
}
```

Test Case Table:

Operation 1: Reverse Each Word

Test Case	Input Sentence	Expected Output
1	What is special?	tahW si ?laiceps
2	Hello Kitty	olleh yttiK
3	Virtual Box	Lautri xoB
4	Luxembourg	gruobmexuL

Operation 2: Print Initials

Test Case	Input name	Expected Output
1	Alexander Graham Bell	A.G.B
2	Nitya Bhasin	N.B
3	Varun Dhawan	A.D
4	Aman Kumar Singh	A.K.S

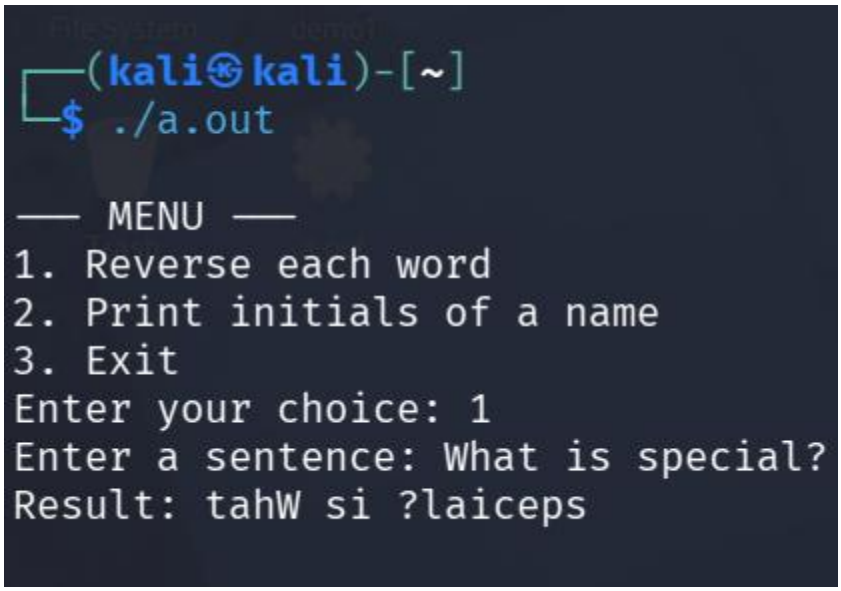
Test Case for the menu:

Test Case	Choice	Expected Output
1	1	Executes reverse-word function
2	2	Executes initials function
3	3	Displays Exiting Program
4	4	Displays Inavalid choice

Compilation and Execution:

The following program has been compiled and executed successfully.

Test Case 1:



Test Case 2:

```
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 1
Enter a sentence: Hello Kitty
Result: olleH yttiK
```

Test Case 3:

```
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: Virtual Box
Enter a sentence: Result: lautri xoB
```

Test Case 4:

```
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 1
Enter a sentence: Luxembourg
Result: gruobmexuL
```

Operation 2:

Test Case 1:

```
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 2
Enter full name: Alexander Graham Bell
Initials: A.G.B.
```

Test Case 2:

```
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 2
Enter full name: Nitya Bhasin
Initials: N.B.
```

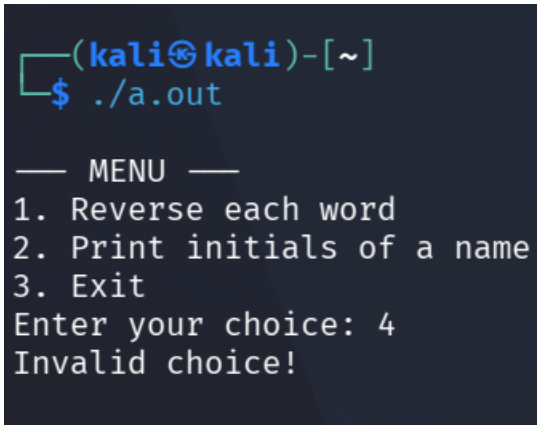
Test Case 3:

```
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: Varun Dhawan
Enter full name: Initials: A.D.
```

Test Case 4:

```
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 2
Enter full name: Aman Kumar Singh
Initials: A.K.S.
```

```
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 3
Exiting program.
```


┌──(kali㉿kali)-[~]
└─$ ./a.out
── MENU ──
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 4
Invalid choice!